

A review of the different approaches to predict the sound transmission loss of building partitions

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Abstract

The computation of sound transmission through solid structures has been of great interest since the first half of the 20th century. Several prediction methods can be found in the large body of literature, applicable to many kinds of structures, and derived using different approaches. This article presents a review of the most significant approaches to compute sound transmission through a building partition; it aims to provide an extensive update and critical literature review of the different approaches which can be used to compute sound transmission through a building partition. Different approaches, suitable for different kinds of building structures, based on analytical and semi-analytical solutions, wave-propagation, numerical or statistical energy analysis methods are described, highlighting advantages and drawbacks.

Keywords

Sound transmission, transmission coefficient, sound transmission loss, prediction models, review

Introduction

The computation of sound transmission through a solid structure has been of great interest to acousticians, at least since the first half of the 20th century. Several analytical and empirical formulations have been derived for many kinds of structures: from simple monolithic homogeneous isotropic thin plates or ribbed panels, to more complex composite systems or double-leaf structures. Moreover, different authors have proposed approximate formulations of equations which otherwise would not have allowed for a closed-form solution. The analysis presented in this article is limited to planar structures, while sound transmission through cylindrical surfaces or curved

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shells is not taken into account. Building partitions involve different technological solutions, spanning from non-homogeneous massive walls to lightweight double-leaf partitions and layered structures, for which more advanced models are required. Nevertheless, the computational power provided nowadays by standard machines allows for an easy numerical implementation of several models which can be used in building acoustic prediction calculations. A comprehensive theoretical background of sound transmission in buildings was given by Heckl¹ almost 40 years ago. More recently, Pellicier and Trompette² published a review of the prediction methods based on the wave approach to compute sound transmission through partitions. The aim of this article is to provide an extensive update and critical literature review of the different approaches which can be used to compute sound transmission through a building partition. Different approaches, suitable for different kinds of building structures, based on analytical solutions, wave-based methods (WBMs), numerical or statistical energy analysis (SEA) methods are described, highlighting advantages and drawbacks. The accurate knowledge of the diffuse incidence transmission loss (TL) of a building partition, which is an important descriptor to evaluate its acoustic performance, also provides the fundamental input data necessary to implement the ISO 12354 models, based on a first-order SEA model,³ which represent the current standard for the prediction of the global acoustic performance in buildings.

Starting from the early works of Cremer, the simple case of a homogeneous, isotropic, single-leaf, infinite, thin panel will be introduced at first. In fact, even though it might be argued that this model has poor applicability to investigate real building partitions, it provides the fundamental knowledge used to derive the theories describing sound transmission through more complex structures. A good agreement between the computed results and the experimental measurements can be achieved only when the model considers the effects of the finite dimension of the structure. This issue will be analysed and investigated in the next section. Ways of including the effects of the diffuse sound field incident on the investigated element are also analysed, since this aspect has a significant influence on the computed results. The analysis is then extended to double-leaf and composite partitions, considering airborne sound transmission via the cavity and structure-borne sound transmission via mechanical connections. Finally, wave propagation-based and numerical approaches are introduced. Such methods allow the modelling of a variety of systems. Thus they can be used to simulate sound transmission through several types of building partitions. In each of the following sections, the most significant bibliographic references for the treated topic will be provided. Moreover, in Appendix 1, approximate formulations and corrections to be applied in the investigated theories are provided.

Single-leaf homogeneous partitions

Sound transmission through single-leaf homogeneous thin panels was thoroughly investigated by Cremer⁴ in 1942. Part of this article, which is in German, and a precis of it in English have been republished by Northwood⁵ and the theory has been described by several authors in English in several textbooks and papers.⁶⁻⁸ If an infinite plane sound wave, with an angular frequency ω and a root mean square sound pressure $p_i(\omega, \theta, \phi)$ is incident on a single-leaf-planar panel mounted in an infinite planar rigid baffle at an incidence angle of θ to the normal to the panel and at an azimuthal angle of ϕ , a reflected plane wave will arise on the incident side $p_r(\omega, \theta, \phi)$, while a plane wave will be radiated on the opposite side by the excited panel, where this latter wave is the transmitted sound pressure $p_t(\omega, \theta, \phi)$. The sound transmission coefficient is defined as the ratio of the transmitted to the incident sound power, even though it can equivalently be computed as the square of the ratio of sound pressure, or alternatively as the ratio of sound intensity. By following the derivation, which can be found clearly explained by the use of different approaches in many

textbooks,^{6,7,9} the transmission coefficient of the partition mounted in an infinite planar baffle separating two acoustic media with equal characteristic impedances can be expressed as

$$\tau(\omega, \theta, \phi) = \frac{\operatorname{Re}\{z_R(\omega, \theta, \phi)\}}{\left|z_R(\omega, \theta, \phi) + \frac{1}{2}z_p(\omega, \theta, \phi)\right|^2 \cos \theta} \quad (1)$$

where $z_p(\omega, \theta, \phi)$ is the normalised impedance that the panel presents to the incident sound wave described above and $z_R(\omega, \theta, \phi)$ is the normalised radiation impedance of the panel when excited by the incident sound wave described above. These two impedances have been normalised by dividing them by the characteristic impedance of the acoustic medium, which is assumed in this case to be the same on both sides of the panel and baffle. This characteristic impedance is equal to the product of the speed of sound in the acoustic medium c_0 and the ambient density of the acoustic medium ρ_0 . All quantities in this article which have dimensions are expressed in the International System of Units. For an infinite homogeneous isotropic panel, by assuming the validity of the thin plate theory, that is, shear deformation and rotatory inertia can be neglected, and the normalised wave impedance can be expressed as

$$z_p(\omega, \theta, \phi) = z_{p,\infty}(\omega, \theta) = \frac{2\omega\rho h}{2\rho_0 c_0} \left[j(1 - r^2 \sin^4 \theta) + \eta r^2 \sin^4 \theta \right] \quad (2)$$

where $r = \omega / \omega_c$ is the ratio between the angular frequency ω of the impinging wave and the angular frequency associated with the first coincidence ω_c (critical frequency), which is given by

$$\omega_c = c_0^2 \sqrt{\frac{\rho h}{D}} \quad (3)$$

The bending stiffness of the plate D is computed from its elastic modulus E and its Poisson's ratio ν

$$D = \frac{Eh^3}{12(1-\nu^2)} \quad (4)$$

where ρ , h and η are the plate's density, thickness and loss factor, respectively. The real part of the normalised radiation impedance of the panel, appearing in the numerator of equation (1), represents the radiation efficiency of the panel. For an infinite homogeneous plate, the normalised radiation impedance is given by

$$z_R(\omega, \theta) = z_{R,\infty}(\theta) = \frac{1}{\cos \theta} \quad (5)$$

Under these assumptions, by substituting equations (2) and (5) into equation (1), the oblique incidence transmission coefficient can be rewritten as

$$\tau_{\infty}(\omega, \theta) = \frac{1}{\left| 1 + \frac{\rho h \omega}{2 \rho_0 c_0} \left[j(1 - r^2 \sin^4 \theta) + \eta r^2 \sin^4 \theta \right] \cos \theta \right|^2} \quad (6)$$

To predict the diffuse field sound transmission coefficient $\tau_d(\omega)$, the incidence and azimuthal angles dependent transmission coefficient $\tau(\omega, \theta, \phi)$ is multiplied by the incident sound intensity, which is proportional to $\cos \theta$ and integrated over all possible solid angles of incidence to obtain the total transmitted intensity. This is then divided by the total incident sound intensity, which is obtained by integrating the incident sound intensity over all possible solid angles of incidence

$$\tau_d(\omega) = \frac{\int_0^{\pi/2} \int_0^{2\pi} \tau(\omega, \theta, \phi) \cos \theta \sin \theta d\theta d\phi}{\int_0^{\pi/2} \int_0^{2\pi} \cos \theta \sin \theta d\theta d\phi} \quad (7)$$

This equation does not allow for a closed analytical solution, even though, under several assumptions and using some simplifications, Cremer⁴ derived approximate solutions of equation (7) for different frequency ranges. Moreover, Davy⁸ extended Cremer's model: removing the greatest part of the original assumptions and replacing the infinite panel radiation efficiency with one for a panel of finite dimension, as reported in Appendix 1. However, for a practical numerical implementation, equation (7) can be discretised, resulting in a weighted sum over a convenient number of angles θ , performed by following a quadrature formulation, such as for example the trapezoidal, or Simpson's rules, or Gauss point integration techniques. It is worth investigating the integration limits of equation (7), since it can be found in several literature references that a better agreement with experimental results would be found by limiting the integration of equation (7) at a value $\theta_l < \theta_{max} = \pi/2$ rad. As discussed by Sharp,¹⁰ it was empirically found that there was a good agreement between the computed transmission loss and the experimental results measured in different laboratories when $78^\circ \leq \theta_l \leq 85^\circ$, often defined as field incidence. This empirical correction allows for a better approximation of the sound transmission below the first coincidence frequency (i.e. the critical frequency), which is governed by non-resonant transmission. A general physical justification of such a limitation has often been the finite dimension of the partition tested in the laboratory^{8,10} which results in a reduced excitation near grazing incidence. However, such an empirical restriction, even though it provides a good approximation of the experimental results, has always been considered unsatisfactory.¹¹

Equation (6) predicts that total sound transmission occurs at grazing incidence when the incidence angle θ equals $\pi/2$ rad. This is not correct for finite-size panels. Below the first coincidence frequency, there does not exist an angle for which the free structural wavenumber can match the acoustic wavenumber: sound radiation and sound transmission, due to the reciprocity of these two phenomena, are governed by forced vibration and modal resonances. For a finite-size isotropic rectangular panel, a finite-size panel normalised radiation impedance z_R needs to be used in equation (1), which, unlike $z_{R,\infty}$, remains finite at grazing incidence when the incidence angle θ equals $\pi/2$ rad. The wave impedance of a finite-size isotropic rectangular panel z_p depends on the dimensions, shape and boundary conditions of the panel and the azimuthal angle ϕ . However, when averaged over a band of frequencies, like a third octave band, the infinite panel normalised impedance $z_{p,\infty}$ is a good approximation. An empirical correction to equation (2) was recently proposed by Davy et al.,¹² in order to predict the directivity of the sound radiated from an

acoustically excited finite thin isotropic panel, as reported in Appendix 1. Therefore, under this assumption, the transmission coefficient of a finite structure τ_{finite} can be computed from the transmission coefficient of the laterally infinite structure τ_{∞} as

$$\tau(\omega, \theta, \phi) = \tau_{\infty}(\omega, \theta, \phi) \operatorname{Re}\{z_R(\omega, \theta, \phi)\} \cos \theta \quad (8)$$

This ignores the change in radiation loading on both sides of the finite panel. The real part of the radiation impedance is the radiation efficiency, for airborne excitation, of the finite-size panel. For a given angular frequency ω , the normalised radiation impedance for a rectangular plate of surface S , depending on the incidence angle θ and the azimuthal angle ϕ is defined as

$$z_R(\omega, \theta, \phi) = \frac{j\omega}{Sc_0} \iint_S \int_S e^{-jk_t(x\cos\phi + y\sin\phi)} G(x, y; \bar{x}, \bar{y}) e^{jk_t(\bar{x}\cos\phi + \bar{y}\sin\phi)} dx dy d\bar{x} d\bar{y} \quad (9)$$

where G represents the half-space Green's function $G(x, y; \bar{x}, \bar{y}) = e^{jk_0 R} (2\pi R)^{-1}$, with $R = \sqrt{(x - \bar{x})^2 + (y - \bar{y})^2}$. Even though this integral does not allow for a closed-form solution, it is possible to find in the literature different approaches to evaluate the finite-size radiation impedance. In fact, in the last 50 years, several authors have proposed formulations to calculate the sound power radiated by a plate of finite dimensions due to the forced vibration caused by airborne excitation. Sewell¹³ derived a solution to take into account the forced sound transmission below the critical frequency. Sato¹⁴ presented tabular and graphical results of the non-resonant (forced) radiation efficiency. Approximate analytical solutions, in good agreement with Sato's results, were developed by Ljunggren¹⁵ and by Davy and colleagues.^{16,17} Thomasson^{18,19} used a variational approach to derive a specific radiation impedance to take into account the influence of the finite geometry of the panel. Thomasson's formulation requires that the panel's aspect ratio is not far from a square one. A similar approach was also used by Brunsog²⁰ to investigate the forced sound transmission through a single-leaf partition. He also discussed the influence of fluid loading in the frequency range near the coincidence frequency. Nowadays, one of the most used approaches was developed by Villot et al.²¹ based on a spatial windowing technique. This approach considers the effect of the diffraction caused by the finite size of the panel inserted into a baffle, by applying a spatial window both to the vibration velocity field, excited on the infinite panel, and incorrectly to the incident sound pressure field. This error in the spatial windowing technique was corrected both by Ghinet and Atalla²² and by Villot and Guigou-Carter.²³ Based on the spatial windowing technique, more computationally efficient formulations to calculate the non-resonant radiation efficiency or the finite-size radiation impedance were developed, for example, by Rhazi and Atalla.²⁴ Moreover, Vigran²⁵ and Bonfiglio et al.²⁶ derived formulations which require even less computational effort, which are valid for a finite-size isotropic rectangular panel that is not too different from a square shape, with an aspect ratio not exceeding 2. In this case, the normalised radiation impedance does not change much when the azimuthal angle is varied and is often approximated by being replaced with its azimuthal average value $z_R(\omega, \theta, \phi) = z_R(\omega, \theta)$. In order to increase the prediction accuracy in the low frequency range, other methods were proposed. For example, in the method proposed by Crocker and Price,²⁷ the resonant contribution, due to the dynamic response of a finite-size panel undergoing free bending vibration, and the non-resonant contribution, due to the forced vibrational field induced in the panel, are considered separately. Wang²⁸ derived the equations to compute the random incidence sound transmission coefficients, based on a modal

summation solution, by introducing an equivalent modal radiation impedance. Although in previous studies simplified modal formulations, which neglected the inter-modal coupling, were used to investigate the sound transmission loss of building elements,²⁹ Wang's investigation highlighted that below the first coincidence frequency this coupling has a significant influence on sound transmission even with light fluids.

Discussion on the assumptions

In the first part of this section, the general theory to compute the sound transmission coefficient of a rectangular thin isotropic plate inserted into an infinite baffle was given, and the equations or the appropriate references to implement a computational code were provided. Even though complete derivations were omitted, different assumptions were made which it is worthwhile to discuss. It can be observed from equation (2) that the wave impedance of an infinite homogeneous isotropic plate is independent of the azimuthal angle ϕ , while it would be necessary to take into account such dependency of the bending stiffness of an orthotropic plate. In fact, anisotropic or orthotropic materials and structures, unlike isotropic ones, have different elastic properties along different directions. A complete derivation of the equations which describe the bending motion of orthotropic plates can be found in several textbooks.^{30,31} In some cases, the spatial in-plane variation of the bending stiffness of orthotropic plates can be approximated using an elliptic function. This approach was used, for example, by Atalla³² to compute sound transmission loss of orthotropic sandwich panels or by Santoni et al.^{33,34} to evaluate both the transmission loss and the radiation efficiency of cross-laminated timber (CLT) partitions. For a wall which is only slightly orthotropic, use of the geometric mean of the bending stiffness is often acceptable. However, for highly orthotropic panels, like corrugated steel panels, the theories of Heckl³⁵ and Hansen^{36,37} can be used.

Kirchhoff's theory for thin plates is only valid as long as the structural wavelength is much larger than the plate thickness. For practical purposes, it is often assumed that the thin plate theory is still valid as long as the structural wavelength is at least six times the plate's thickness: $\lambda_{st} \geq 6h$, even though such a limit was never mathematically proven. As the plate's thickness becomes comparable with the structural wavelength, the rotatory inertia and shear deformation start significantly influencing the sound transmission. The latter in particular limits the otherwise continuous increase with frequency of the pure bending wave velocity. The structural wave propagating in an infinite thick plate can be seen as a combination of longitudinal and shear waves, which tends to a pure bending wave only as the frequency tends to zero and which gradually shifts to a shear wave as the frequency increases. Flexural wave propagation in thick plates can be described by means of Mindlin's theory, as outlined by Rindel.³⁸ In his paper Rindel presented an experimental approach to evaluate the phase velocity of structural waves propagating in a thick panel, providing the dispersion relation for a flexural wave under Mindlin's assumption for thick plates, as originally proposed by Kurtze and Watters.³⁹ This experimental method was also applied by Santoni et al.⁴⁰ in order to investigate the propagation of structural waves in CLT panels. In the transmission loss spectrum, these effects are generally highlighted by a plateau region at the higher frequencies, caused by thickness resonances occurring when the half-wavelength of the structural compressional wave equals the wall thickness. Ljunggren^{41,42} derived an analytical expression to compute sound transmission loss of arbitrary thick plates, providing more accurate results compared to the solution based on Mindlin's approximate theory for thick plates, which was derived by Heckl and Donner⁴³ for an infinite plate.

Double-leaf partitions

The analysis of sound transmission through double-leaf structures is certainly more complex than the theory outlined in the previous section. A great variety of models were proposed by several authors during the last 80 years. Among these, the most referenced were described and compared by Hongisto⁴⁴ in a thorough and critical review of the prediction approaches applicable to double-leaf building walls. Due to the complexity of the phenomenon, most of these methods are based on several assumptions, which make them applicable only to very specific conditions. The approach derived by Beranek and Work⁴⁵ in 1949, which only considers normal sound incidence excitation, together with the model proposed by London⁴⁶ in 1950, which is limited to empty cavities, is often referred to as the classical model. The model proposed by Beranek and Work, based on the impedance transfer method, is theoretically rigorous and allows the investigation of several double-leaf and cavity structures. However, it was derived only for the case of normal incidence sound transmission, thus it can be seen to be of poor practical use. Mulholland et al.^{47,48} extended this model for the case of an arbitrary angle of incidence θ of the impinging acoustic plane wave. The sound transmission coefficient of an infinite double-leaf wall, consisting of an air cavity with thickness d between two panels with surface mass m_1 and m_2 , can be expressed as⁷

$$\tau(\omega, \theta) = \left| \frac{2j \sin(k_0 d \cos \theta) \sec^2 \theta}{z'_{p,1} z'_{p,2} \sin^2(k_0 d \cos \theta) + \sec^2 \theta} \right|^2 \quad (10)$$

where $z'_{p,i} = z_{p,i} + \sec \theta [1 - j \cot(k_0 d \cos \theta)]$, indicating with $z_{p,i}$ the normalised specific wave impedance of the i th leaf. A double-leaf cavity wall has a normal incidence mass-air-mass resonance at an angular frequency of

$$\omega_n = c_0 \sqrt{\frac{\rho_0 (m_1 + m_2)}{d m_1 m_2}} \quad (11)$$

Below ω_n , the two wall leaves are effectively rigidly coupled, and the sound insulation is the same as that of a single-leaf wall with the same total mass per unit area as the double-leaf cavity wall. For an infinite double-leaf cavity wall, for angular frequencies at and above ω_n , there is always an angle θ_0 at which an oblique incidence mass-air-mass resonance occurs with a resonance frequency $\omega_{n,0} = \omega_n / \cos \theta_0$. If there is no sound-absorbing material in the air cavity of an infinite double-leaf cavity wall, London⁴⁶ showed that its sound transmission coefficient above ω_n is actually greater than the sound transmission coefficient of a single-leaf wall with the same total mass per unit area. This is not the case for finite double-leaf cavity walls because, as Cambridge et al.⁴⁹ showed in this case, in any finite frequency range, there are only a finite number of discreet angular frequencies and angles of incidence at which an oblique incidence mass-air-mass resonance occurs. This is because there must be resonances along the length and along the height of the cavity as well as across the cavity in order to obtain an actual resonance. The presence of sound-absorbing material in the cavity of a double-leaf wall effectively damps out the resonances along the length and height of the cavity and hence damps out the oblique mass-air-mass resonances. This enables the infinite double-leaf cavity wall theory to be used to predict the sound transmission coefficient of a finite double-leaf cavity wall with sound-absorbing material in the wall cavity. To predict the sound transmission coefficient of a finite double-leaf cavity wall, with no sound-absorbing material in the air cavity, a fictitiously large value of the attenuation coefficient, or sound

absorption of the air cavity, needs to be used with the infinite double-leaf cavity wall theory. This is necessary in order to artificially account for the fact that there are only a finite number, rather than an infinite number, of lightly damped oblique mass-air-mass resonant modes. The kind of material used to fill the cavity and its thickness have a significant influence on the mass-spring-mass resonance and on the value of the TL above this frequency. As a first approximation, the elastic frame of the porous material, assuming this is not in contact with the partitions or cannot set in motion the partitions due to vibration of its solid structure, can be considered as a rigid and motionless skeleton, while the air filling the interconnected pores is described as an equivalent fluid medium. Such an equivalent fluid is characterised by a complex density and a complex bulk modulus which give the complex wavenumber and the complex characteristic impedance.⁵⁰ In the literature, several empirical models can be found to describe porous and fibrous materials as an equivalent fluid, requiring the knowledge of a certain number of physical properties.^{51–55} However, for some porous materials, it is necessary to also consider the skeleton deformation, that is, its elasticity. Biot's^{56,57} theory can be used to evaluate sound propagation through a porous material considering the interaction between its elastic frame and the fluid filling the pores.

Sound transmission through structural connections

In real structures, the two leaves of double-leaf partitions are mounted onto a frame. Even though the presence of such a mechanical connection has been neglected so far in this article, it has a significant influence on the sound transmission loss of a wall, causing a reduction of sound insulation at high frequencies, due to an additional structure-borne sound transmission path via the connections.⁵⁸ A decoupled approach to take into account sound transmission via the structural links in double-leaf panels was developed in 1978 by Sharp,¹⁰ based on the work of Cremer et al.⁶ for frequencies less than about half the lowest critical frequency of the wall leaves, and by using Heckl's⁵⁹ formulae for the near-field radiation from point and line connections. In this theory, the airborne and the structure-borne sound paths are considered independent and separately computed. The transmission coefficient of the partition can be expressed as

$$\tau_{tot} = \tau \left(1 + \frac{W_b}{W_a} \right) \quad (12)$$

where W_b is the sound power radiated due only to the action of sound bridges, while W_a represents the sound power that would be radiated by the structure without the presence of structural connections. While the model derived by Sharp only allows for rigid and massless structural links, Gu and Wang⁶⁰ introduced the possibility to consider resilient connections. The assumption of perfectly rigid connections overestimates the reduction of sound insulation at higher frequencies, providing results lower than what would be determined experimentally on double-leaf partitions with steel studs manufactured from thin sheet steel. Sharp's decoupled approach was extended up to the frequency region near and above the critical condition by several authors, following different approaches. Davy⁶¹ removed an empirical correction used by Sharp, by including the resonant vibration of the wall leaf on the excited side and the radiation of the resonant vibration of the wall leaf on the receiving room side; he also extended Sharp's theory so that it can also be applied above half the critical frequency. In Davy's theory, the structure-borne sound transmission between the wall leaves due to point- or line-connections is modelled as four pole networks.^{62,63} The connections can be considered either as massless and rigid links or resilient and with a finite mass. For example, Davy's⁶¹ formulation to compute the diffuse incidence sound transmission coefficient of

a double-leaf partition due to structural connections between the wall leaves, with n resilient massless point connections in a unit area, is

$$\tau_d(\omega) = \frac{256n\rho_0^2 c_0^4 QR}{\pi\omega^2 \left[(m_1\omega_{c2} + m_2\omega_{c1})^2 + 64\omega^2 m_1^2 m_2^2 c_0^2 C^2 \right]} \quad (13)$$

while for uniformly spaced parallel line connections, at a distance b of one another, equation (13) becomes

$$\tau_d(\omega) = \frac{64\rho_0^2 c_0^3 QR}{b\omega^2 \left[G^2 + (4C\omega^{3/2} m_1 m_2 c_0^2 - G^2) \right]} \quad (14)$$

where

$$G = m_1 \sqrt{\omega_{c2}} + m_2 \sqrt{\omega_{c1}}; \quad Q = \begin{cases} 1+e, & \text{if } \omega < \omega_{c1}; \\ e, & \text{if } \omega \geq \omega_{c1}; \end{cases} \quad e = \frac{\pi\omega_{c1}\sigma_1}{4\omega\eta_1};$$

$$R = \begin{cases} 1+r, & \text{if } \omega < \omega_{c2}; \\ r, & \text{if } \omega \geq \omega_{c2}; \end{cases} \quad r = \begin{cases} \frac{\pi\omega_{c2}\sigma_2}{4\omega\eta_2} & \text{for point connections;} \\ \frac{\sigma_2}{2\eta_2} \sqrt{\frac{\omega_{c2}}{\omega}} & \text{for line connections.} \end{cases}$$

The term C represents the compliance, that is, the inverse of stiffness, of the point or line connections, which, in the first case, has the dimensions length/force, in the latter of 1/pressure. The subscript 1 refers to the wall leaf on the sending room side and the subscript 2 refers to the wall leaf on the receiving room side. Unlike equation (13), equation (14) is not symmetrical with regard to wall leaf number. As explained in detail by Davy,⁶¹ this happens because it is possible for total internal reflection to occur for vibration incident at a non-normal angle to a line connection. More recently, Hirakawa and Davy⁶⁴ have obtained empirical formulae for the translational compliance of resilient channel bars and of thicker gauge steel studs. Vigran⁶⁵ developed a modified decoupled approach within the transfer matrix method (TMM), which allows to take into account the structure-borne sound transmission due to point and line connections. Vigran⁶⁶ introduced the possibility to consider studs with finite stiffness, by fitting a third-order polynomial to the numerical results obtained by Poblet-Puig et al.,⁶⁷ who modelled the frequency-dependent stiffness of thin gauge steel studs. This method does not include the transmission due to resonant vibration below the critical frequency. The magnitude of the error in neglecting the resonant transmission below the critical frequency depends on the magnitude of the damping loss factors of the two wall leaves. The theoretical formulation to compute the mass-spring-mass resonance frequency, given in equation (11), provides accurate results for double-leaf walls without mechanical bridges, or for cavity partitions with highly resilient structural connections. However, in cavity walls with more rigid studs, it can be experimentally observed that the mass-spring-mass resonance is significantly shifted towards higher frequencies compared to the one that would be predicted using equation (11). An increase in the mass-air-mass resonance frequency was theoretically predicted by Lin and Garrelick⁶⁸ and

experimentally observed by Bradley and Birta⁶⁹ in the laboratory although not in the field. Davy et al.⁷⁰ have recently explained that the increase in the mass-air-mass angular resonance frequency in two leaf cavity stud walls is due to the spring of the air in the wall cavity coupling, not the masses per unit area of the wall leaves, but the drum modes of the wall leaves between the studs. Other approaches can be found in the literature to compute structure-borne sound transmission through the mechanical connections of double-leaf partitions, which cannot all be described in detail in this article. However, among them, it is worth mentioning the approach proposed by Craik and Smith,^{71,72} in which the airborne and structure-borne sound transmission through double-leaf walls is computed within the SEA framework, and the methods based on the assumption of infinitely periodic structures.^{62,68} All the mentioned models assume a single layer of connections and the validity of thin plate theory for the leaves of the partition. Santoni et al.⁷³ extended Vigran's model within the TMM framework to any three-dimensional (3D) elastic solid. Moreover, Chevillotte and Marchetti⁷⁴ developed a model which also allows to consider multiple layers of studs.

Sandwich panels

Sandwich panels, representing a variation of double-leaf structures, are probably not as widely used in building construction as in the aerospace, automotive or naval industries. It is beyond the scope of this article to present an extensive analysis of the literature related to sandwich panels. However, since they represent a fundamental solution in noise control engineering, it is appropriate to mention the basic references which present the background theories implemented in the algorithms currently used. A sandwich panel generally consists of two thin laminates bonded by a thick and lightweight core. Kurtze and Watters³⁹ began investigating sandwich panels in order to identify solutions with an increased acoustic performance but still able to fulfil structural stability requirements, providing a model to predict sound transmission through this kind of element. The theory derived by Kurtze and Watters was extended by several authors. Guyader and Lesueur^{75,76} studied oblique and diffuse sound transmission through orthotropic multilayer elements, while Nilsson⁷⁷ provided a general description of the bending motion of sandwich structures. The laminae, the outer layers of a sandwich elements, are generally treated as thin plates, while the core motion is described by taking into account shear deformation and rotary inertia. A peculiarity of these structures is an apparent frequency-dependent bending stiffness.⁷⁸ At very low frequencies, the dynamic response of a sandwich beam is governed by its static bending stiffness D_s , depending on the elastic moduli of the core and of the laminae and their associated thicknesses. As the frequency increases, the apparent bending stiffness decreases asymptotically approaching the bending stiffness of the laminae D_l . The assumption of the laminae moving in phase is valid below the mass-stiffness-mass or dilatation resonant frequency

$$f_d = \frac{1}{2\pi} \sqrt{\frac{E_c (\rho_1 h_1 + \rho_2 h_2)}{h_c \rho_1 h_1 \rho_2 h_2}} \quad (15)$$

where E_c is the elastic modulus of the core and h_c its thickness, while ρ_i and h_i are the density and thickness of the laminae. Using apparent frequency-dependent elastic properties, both for isotropic and orthotropic panels, it is possible to calculate the sound transmission coefficient of a sandwich plate using the equation derived in the previous section for a homogeneous single-leaf wall.^{79–82} Because sandwich panels are lightweight and stiff, they have low sound insulation at low frequencies, and also at middle frequencies, since, due to their high bending stiffness, they also

exhibit a low critical frequency. If the core of a sandwich panel is not stiff, the shearing of the core produces an effective bending stiffness and hence a critical frequency which varies with frequency. A review of theoretical models to predict the acoustic performance of sandwich panels was recently published by D'Alessandro et al.⁸³

Sound transmission through complex structures

Building partitions are often composite systems constituted by several materials and multiple layers. For example, traditional masonry partitions are inhomogeneous and anisotropic structures, made with clay or concrete bricks coupled together by mortar joints. Moreover, both traditional partitions and lightweight systems, such as CLT panels, are generally plastered on one or both sides, or clad in order to achieve a better sound and thermal insulation. The models described so far, for homogeneous single and double-leaf walls, are not suitable to compute sound transmission through such complex structures. In the next sections, numerical and wave propagation-based approaches suitable to investigate layered or non-homogeneous media are described.

Transfer matrix method

The sound transmission coefficients of infinite lateral extent multilayer walls for a particular angle of incidence can be predicted using the transfer matrix method (TMM). It is one of the most computationally efficient and most widely used methods to describe the acoustic wave propagation through layered structures.^{84–86} The TMM solves a two-dimensional (2D) problem of a plane acoustic wave impinging on the surface (S_1) of the layered element, as sketched in Figure 1. The general formalism of TMM can be expressed as

$$V_{(S_1)} = [T] V_{(S_2)} \quad (16)$$

All the variables defining the acoustic field on the surface (S_1) are given by the vector $V_{(S_1)}$, while the vector $V_{(S_2)}$ contains the descriptors related to each single layer and all the variables that define the acoustic fields on the surface (S_2), such as pressure, velocity and stresses. The transfer matrix $[T]$ describes the sound transmission through the layered structure. The size of this matrix depends on the number of layers and on their nature. In fact, this method allows the consideration of many different media: elastic solids, either isotropic or anisotropic,⁸⁷ viscoelastic materials and thin plates. In addition, it is possible to take into account porous materials, either described as equivalent fluids^{51–54} or by means of Biot's theory to consider wave propagation both in the elastic frame and in the fluid filling the open pores, which characterises acoustical poro-elastic materials. A clear description of Biot's theory of sound propagation in isotropic porous media can be found in Chapter 6 of Allard and Atalla.⁵⁰ Moreover, Bécot and Jaouen⁸⁸ derived Biot's equations for any acoustic model. It is also possible to account for the anisotropy of porous materials.^{89,90} Besides, as shown by Atalla and Sgard,⁹¹ also perforated plates and screens can be easily modelled as a porous medium. The TMM method offers an easy implementation, described in detail in Chapter 11 of Allard and Atalla,⁵⁰ and has a wider applicability compared to the analytical approaches previously described, since it allows the modelling of composite systems made of a variable number of layers, considering media of different natures. The sound transmission coefficient can be (numerically) integrated over all possible solid angles of incidence to obtain the diffuse field transmission coefficient. For finite lateral extent multilayer walls, a correction for the difference between the infinite

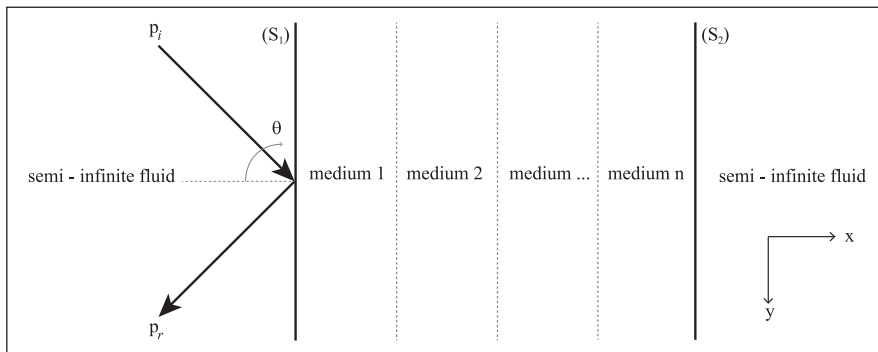


Figure 1. TMM diagram: plane wave incident with angle θ on a multilayer structure surrounded by a semi-infinite fluid on both sides.

and the finite radiation efficiencies must be made. The spatial windowing technique previously described, which effectively calculates the finite radiation efficiency, can also be used. It should be mentioned that the TMM framework allows for the computation of several vibro-acoustic parameters and quantities, the consideration of different kinds of excitation other than acoustic plane waves, such as mechanical forces, and turbulent boundary layer excitation.^{32,92,93} Besides, it is also possible to take into account the structure-borne sound transmission through the mechanical connections in double-leaf walls and partition cladding systems.^{65,73}

Numerical and hybrid methods

The TMM approach certainly represents a powerful tool to evaluate the sound insulation performance of multilayer structures, which is particularly suitable to predict the vibro-acoustic behaviour of building partitions, however, it might be argued that it provides a good approximation in the mid- to high-frequency range but lacks accuracy at lower frequencies when the investigated element exhibits a strong modal behaviour. Theoretically, the finite element method (FEM) and the boundary element method (BEM)⁹⁴ could be used in order to accurately compute the vibro-acoustic behaviour of a great variety of structures including fluid, elastic and poro-elastic media,⁹⁵ regardless of the complexity of their geometry, or the inhomogeneity of the materials involved, and also by taking into account the coupling between the partition and both the source and receiving rooms. A number of examples of numerical calculation to investigate the effects on the sound transmission of the rooms or of the finite size of the partition, both of which might be significant at the low frequencies, can be found in the literature.^{96–98} Davidsson et al.⁹⁹ investigated sound transmission through a lightweight partition between two rooms by means of a structural-acoustic finite element (FE) analysis, in 1/3 octave bands between 40 and 200 Hz. Brunskog and Davidsson¹⁰⁰ proposed a more efficient FEM approach to investigate the sound insulation of a lightweight partition at low frequencies, by considering the sending and the receiving rooms as waveguides. However, this model was applied in a limited frequency range from 50 up to 630 Hz. A numerical analysis within the entire building acoustics frequency range (100–3150 Hz) of sound transmission through a double-leaf wall by means a 3D FE model, which also considered fluid-structure coupling, was presented by Arjunan et al.¹⁰¹ Jean et al.¹⁰² investigated sound transmission through massive partitions by means of a vibro-acoustic decoupled approach, in which the dynamic response of the structure

was computed using a FE model while the pressure fields in the sending and receiving rooms were evaluated by means of a decoupled modal approach. Santos and Tadeu^{103,104} investigated in two dimensions the problem of sound transmission through a single-leaf partition using the BEM. Nevertheless, despite the great effort in reducing the computational cost of these numerical methods and the rapid development of powerful microprocessors, such approaches remain computationally expensive and can thus be applied in a very limited frequency range, compared to the one of interest in building acoustics. On the other hand, several successful attempts have been made in order to develop hybrid solutions, combining numerical methods, which can provide a detailed description of the elastic and the acoustic domains, with more computationally efficient approaches. Rhazi and Atalla⁹³ developed a modal-based TMM to evaluate the vibro-acoustic behaviour of layered structures, whose dynamic response, due to various types of excitation, is described as a sum of modal contributions. Alimonti et al.^{105,106} proposed a hybrid FE-TMM approach to evaluate the vibro-acoustic descriptors of flat structures, combining the accuracy and versatility offered by FEMs with the efficiency of the TMM, in order to significantly reduce the computational effort. Van den Wyngaert et al.¹⁰⁷ employed a deterministic-SEA hybrid approach to couple the analytical model of wall leaves and the FE model of the metal frame with the diffuse sound field in the emitting and receiving room, in order to determine the sound insulation of a double-leaf wall with a flexible frame. This approach, consisting in a deterministic description of the partition coupled by means of power balance to a stochastic modelling of the rooms, was previously used to evaluate sound transmission through different kinds of building partitions by Reynders et al.¹⁰⁸ The WBM¹⁰⁹ was used by Dijckmans et al.¹¹⁰ to compute the sound transmission loss of cavity walls by taking into account coupling between the rooms, the panels, and the air cavities. Beside, Dijckmans and Vermeir¹¹¹ presented a hybrid approach combining the WBM with the TMM in order to investigate the vibro-acoustic behaviour of multilayer structures, consisting of elastic and poroelastic materials, again by taking into account the fluid-structure coupling. The WBM is used to evaluate the modal behaviour of both the partition and the two rooms, while the TMM is used in order to compute the coefficients of the modal impedance matrix, assuming infinite layers and an imposed trace wavenumber for each considered structural mode. Numerical models have also been applied in order to investigate inhomogeneous walls, such as masonry partitions, made for example of hollow bricks and mortar joints. Maysenhölder and Haberkern¹¹² implemented numerically an analytical solution to compute sound transmission through periodically inhomogeneous plates, which was derived within the framework of the 3D elasticity theory and by means of a variational approach. A 2D FE model with fluid structure interaction was implemented by Del Coz Diaz et al.,¹¹³ for the horizontal and the vertical plane, in order to investigate sound transmission through a lightweight concrete hollow brick wall. Jacques et al.¹¹⁴ modelled a hollow brick wall as an equivalent homogeneous orthotropic thick wall within the TMM framework, describing it by means of effective elastic properties, which were determined by means of a numerical analysis performed on the single brick block. The wave and finite element (WFE) method¹¹⁵ applies a standard FE modelling on a small area of the structure, with periodic boundary conditions, to calculate its equivalent dynamic properties. Wave-propagation methods are then used to calculate the sound transmission coefficient of the infinite panel.^{116,117} The WFE approach can be used as a valid alternative to the analytical derivation of the transfer matrices characterising the investigated element, generally used in the TMM, which, for complex structures, might be a time-consuming and cumbersome process. Such approach also allows to consider non-planar geometries such as cylindrical structures.¹¹⁸ Moreover, other models were recently developed with the aim of adapting the TMM to curved surfaces.^{119,120}

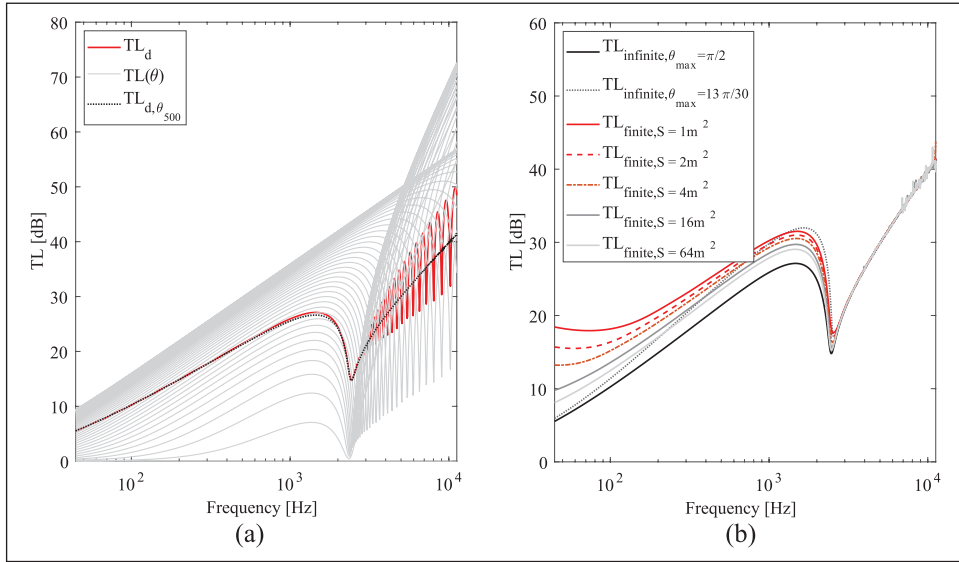


Figure 2. Sound transmission loss of a homogeneous isotropic infinite plasterboard panel: (a) the diffuse incidence TL_d is plotted together with the 45 associated angle-dependent $TLs(\theta)$. The diffuse incidence $TL_{d,\theta_{500}}$ is computed by considering 500 discrete angles within the range $0 \leq \theta_i \leq \pi/2$ rad. (b) Comparison between infinite panels and spatially windowed finite-size panels.

Results and discussion

In this section, results to illustrate differences and similarities between the various approaches which have been discussed in the previous sections are presented. The base structure investigated is a 12.7-mm-thick plasterboard panel, with density $\rho = 650 \text{ kg/m}^3$, elastic modulus $E = 2.8 \text{ GPa}$, loss factor $\eta = 0.01$ and Poisson's ratio $\nu = 0.25$. The oblique incidence sound transmission loss, given by $TL(\theta) = -10 \log \tau(\theta)$, of such a single-leaf panel was computed for a discrete number of angles $0 \leq \theta \leq \pi/2$ rad using equation (1) and is shown in Figure 2(a). For each incidence angle θ , the coincidence condition, at which the trace wavenumber induced by the acoustic plane wave equals the free bending wavenumber propagating in the plate: $k_t = k_0 \sin(\theta) = k_B = (\omega^2 \rho h / D)^{(1/4)}$, can be clearly identified. The lowest angular coincidence frequency is found for grazing incidence $\theta = \pi/2$ rad. In Figure 2(a), the diffuse field transmission loss of the same plasterboard panel, computed by discretising the Paris formulation given in equation (7), is shown. All the dips highlighted by the red solid curve in Figure 2(a), representing the diffuse incidence sound transmission loss TL_d computed considering 45 discrete angles $0 \leq \theta_i \leq \pi/2$ rad, are caused by the coincidence condition associated with each angle of incidence of the impinging plane wave. This numerical effect can be reduced by increasing the number of discrete angles, as shown in the same figure by the dotted black line: $TL_{d,\theta_{500}}$ which represents the diffuse sound transmission loss averaged over 500 angles $0 \leq \theta_i \leq \pi/2$ rad. In Figure 2(b), the diffuse incidence transmission loss computed for homogeneous plasterboard panels, with a varying surface area from $S = 1 \text{ m}^2$ up to $S = 64 \text{ m}^2$, is compared with the TL computed for a laterally infinite panel, either considering the entire range of angles of incidence $0 \leq \theta \leq \pi/2$ rad, or the reduced one with $\theta_{\max} = 78^\circ = (13/30)\pi$ rad. The finite-size correction was applied by following the implementation proposed by Bonfiglio et al.²⁶ The validation of this approach is not among the objectives of this review, since several comparisons

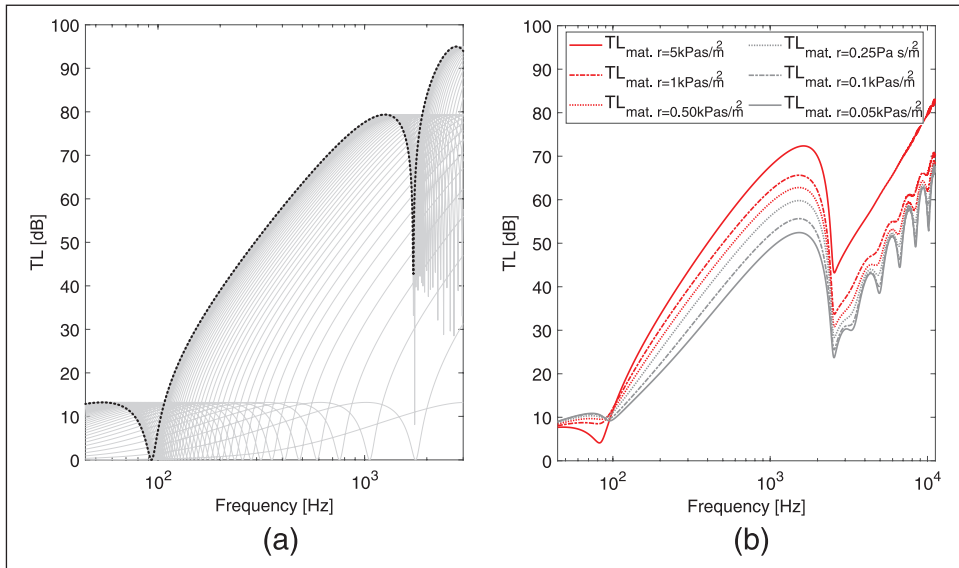


Figure 3. Sound transmission loss of an infinite-size double-leaf wall with two identical panels delimiting a cavity 100 mm thick: (a) angle-dependent TL of the limp wall $TL_s(\theta)$, normal incidence $TL_{\theta=0}$ and (b) diffuse incidence TL of the double-leaf wall (panels' elasticity is considered), computed for different values of the airflow resistivity of the medium filling the cavity.

between numerical and experimental data can be found in the cited papers. However, it can be useful for the reader to understand the influence of the finite-size correction on the computed TL. By comparing the TL curves provided in the graph in Figure 2(b), it is evident that the effect of the spatial windowing is confined to below the critical frequency, which for the considered elements falls around 2500 Hz. Above the critical frequency, all the computed curves are identical. The finite-size correction increases the sound transmission loss below the critical frequency and the correction increases as the frequency decreases, unlike the curve obtained by simply reducing the maximum angle of incidence. Moreover, as the panel surface increases, the finite-size TL approaches the one computed for an infinite panel. According to Wang,¹²¹ the increase in TL as the frequency decreases for very low frequencies and small panel sizes is not correct. This occurs because the fluid loading is not correctly accounted for.

The sound transmission loss of the double-leaf wall, made of two identical plasterboard panels $m_1 = m_2 = 8.255 \text{ kg/m}^2$ delimiting a 100-mm-thick air cavity, was computed using equation (10); it is shown in Figure 3(a) for different values of the incidence angle θ . In this case, the plate's stiffness and damping are neglected, thus the plates' impedance is computed considering only the inertial term: $Z_i = j\omega\rho_0 c_0$. The dashed black line represents the TL computed for normal incidence of the impinging sound wave ($\theta = 0$ rad). The first dip, falling around 92 Hz, occurs at the physical mechanism known as the normal incidence mass-air-mass resonance. This resonance, which strongly influences the performance of double-leaf partitions, is inversely proportional to the surface masses of two plates m_1 and m_2 and to the thickness of the cavity d , as described by equation (15). Above this frequency, the TL of a double-leaf wall is significantly higher than the sound insulation that would be provided by a single-leaf wall with the same surface mass. However, as London showed, for the diffuse field sound transmission loss, this is true only if there is sound-absorbing material filling the cavity. This is because of the infinite number of dips in the TL curves

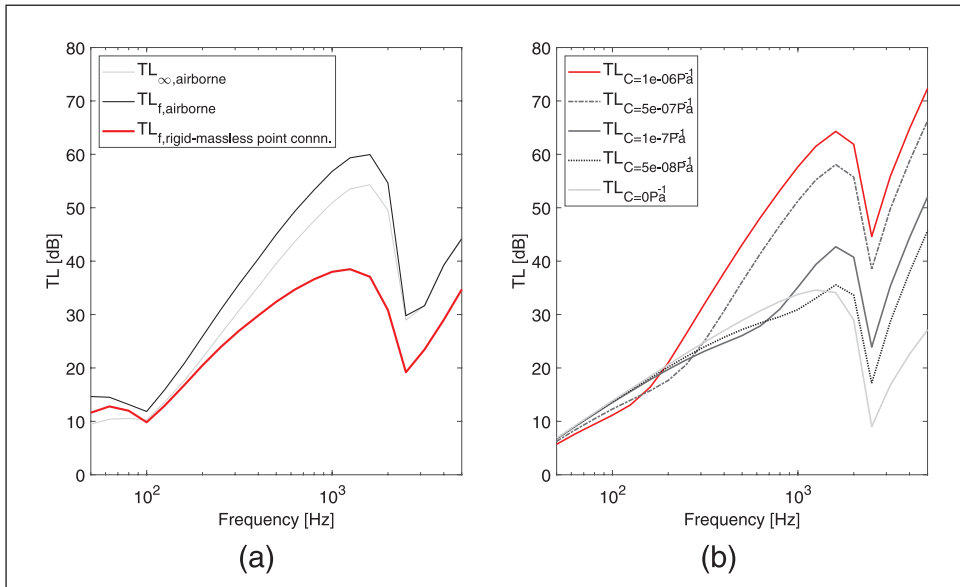


Figure 4. Sound transmission loss of a double-leaf infinite plasterboard wall with two identical panels delimiting an air cavity 100 mm thick. Results presented in 1/3-octave bands: (a) results obtained from a TMM model and (b) results obtained using equation (14) for different values of stud compliance C .

that can occur between the normal incidence mass-air-mass resonance frequency and the first cross cavity resonance. As noted above, for a cavity wall of finite lateral extent, only a finite number of such dips can occur. The normal incidence transmission loss, the black dotted curve indicated by $TL_{\theta=0}$, exhibits a dip above the mass-spring-mass resonance around 1700 Hz, which is associated with the cross cavity resonance. Cross cavity resonances occur at frequencies at which the cavity depth d equals a multiple of half a wavelength. Experimental measurements of diffuse incidence TL in third octave bands show that cross cavity resonances can usually be ignored. Figure 3(b) shows the sound transmission of the same double-leaf partition but with sound-absorbing material filling the cavity and not bonded to the partition leaves, which was computed using the TMM, by modelling the porous material as an equivalent fluid according to the Delany and Bazley model. Different materials, all with a thickness of 100 mm, were simulated by varying the airflow resistivity within the range $r = 0.05 - 5 \text{ (k Pa s/m)}^2$. The presence of the material inside the cavity increases the sound insulation performance of the wall and reduces the dips due to the cavity resonances in the high-frequency range.

Figure 4(a) shows the effect of structural connections on the sound transmission loss, which is reduced due to the action of mechanical bridges. These results were computed using a TMM model which considers infinitely stiff massless point connections.^{65,73} In Figure 4(b), the sound transmission loss of the double-leaf panel is computed considering elastic and massless line connections, according to equation (14) for different values of the stud compliance C : from $C = 1e-06 \text{ Pa}^{-1}$ down to $C = 0 \text{ Pa}^{-1}$ which corresponds to rigid connections. The results in Figure 4 are given in 1/3-octave frequency bands. From this comparison, it is evident how the assumption of perfectly rigid and massless connections overestimates the sound transmission through the resilient structural connections, compared to the case when their elasticity is considered. As the compliance of the connections increases, the effect of structure-borne sound transmission is less and less significant.

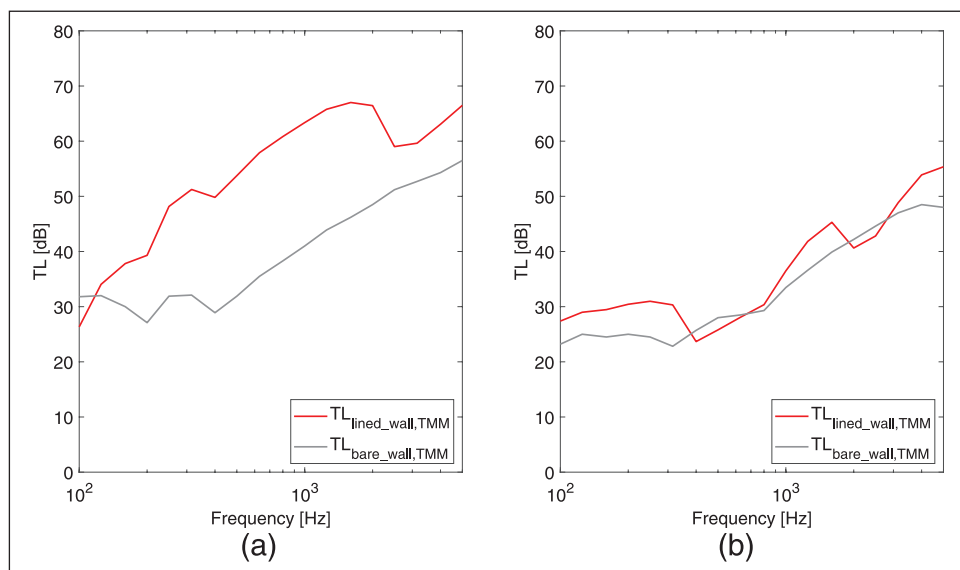


Figure 5. Sound transmission loss of building partitions presented in 1/3-octave bands: (a) clay hollow brick wall with and without lining and (b) 3-ply CLT panel with and without plasterboard lining.

Finally, the sound insulation performance computed for more complex partitions is shown in Figure 5 in 1/3-octave frequency bands. The sound transmission loss of horizontally perforated clay hollow bricks bound with mortar joints and plastered on both sides, with a total thickness of 100 mm, is given in Figure 5(a). Results are shown both for the bare wall, and also for the condition in which this was lined with a single layer of plasterboard, fixed with fasteners to metal studs, and the cavity was filled with glass wool material. The TL was computed using a TMM model, including the finite-size correction and the presence of rigid and massless studs. Figure 5(b) shows the TL computed for a 3-ply CLT panel 78 mm thick. The results are presented both for the bare wall configuration and for the CLT panel lined with a double layer of fire-rated plasterboard, screwed into the wood. All the details regarding the modelling of such partitions and the required input data can be found in the paper recently published by Santoni et al.¹²²

Conclusion

This article presents an extensive analysis of the most used approaches to compute sound transmission through building partitions. Starting from the simplest case of an infinitely extended homogeneous single-leaf panel, the authors provided the basic equations to compute the sound transmission coefficient, based on the wave propagation approach. The influence of the finite dimensions of the panel was thoroughly discussed, describing how the accuracy of the results, below the critical frequency, can be increased by including in the computation the real part of the normalised non-resonant radiation impedance. The authors also analysed double-leaf walls, providing the equation to compute the transmission coefficient, derived with the wave propagation approach. Moreover, the article also provides some references on sandwich panels. The significant influence of the structural connections in double-leaf elements was considered, by describing different methods to account for these effects when computing sound transmission through these kinds of structures. The TMM was introduced as a convenient approach to model multilayer structures. Even though

the methods based on the wave propagation approach were of primary concern, the authors mentioned several works which treated sound transmission through a partition differently, for example, by considering its modal response. They also considered the possibilities offered by numerical methods. Even though, at the moment, a fully coupled vibro-acoustic simulation, within the entire building acoustics frequency range, still requires a too high computational cost for widespread use of these methods, they represent helpful and powerful resources in combination with other approaches. Finally, in the last section, the authors provided some examples of the discussed topics and showed the possibilities given by different approaches by comparing simulated results in terms of sound transmission loss.

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Appendix I

Corrections and approximations

The normalised wave impedance of an infinite plate, given in equation (2), can alternatively be expressed in terms of lumped elements as

$$z_{p,\infty}(\omega, \theta) = 2a \left[j(1-u) + \eta u \right] \quad (17)$$

$$a = \frac{\omega \rho h}{2\rho_0 c_0}; \quad u = r^2 \sin^4 \theta; \quad r = \frac{\omega}{\omega_c} \quad (18)$$

It should be noted that Davy et al.¹² has recently found it necessary to introduce an empirical correction to equation (17) when predicting the directivity of the sound radiated from a finite-size thin isotropic panel which is excited by sound incident from the other side. This empirical correction is the replacement of u with t given by the following equation

$$t = \begin{cases} u, & \text{if } u < t_l \\ t_l, & \text{if } u > t_l \end{cases} \quad (19)$$

The term limit t_l is the upper limiting value of the term t , which was set by Davy to the value of $t_l = 2$. This restricts the magnitude of the imaginary part of the normalised panel impedance to

being less than or equal to that of an infinite limp version of the panel. When $\omega \ll \omega_c$, the normalised wave impedance is mostly governed by the plate inertia and equation (17) can thus be approximated as

$$z_{p,\infty}(\omega, \theta) \approx z_{p,\infty}(\omega) = j2a \quad (20)$$

Except for low frequencies or lightweight panels, the term $a[j(1-u) + \eta u]$ is usually very much greater than the normalised radiation impedance z_R ; therefore, an approximation is often made by ignoring the term z_R in the denominator of equation (1). By considering this, together with the approximation expressed in equation (20), it is possible to approximate equation (8) as

$$\tau(\omega, \theta) \approx \frac{\text{Re}\{z_R(\omega, \theta)\}}{a^2 \cos \theta} \quad (21)$$

Introducing this approximation into equation (7), the diffuse incidence sound transmission coefficient of a finite-size panel well below the coincidence angular frequency ω_c is given by

$$\tau_d(\omega) \approx \frac{2}{a^2} \langle \sigma \rangle \quad (22)$$

where the angular brackets $\langle \rangle$ denote that the forced radiation efficiency σ , as already mentioned, is the real part of the normalised radiation impedance and is averaged over all possible solid angles. Since there are several good approximations for the average radiation efficiency $\langle \sigma \rangle$, equation (22) is a good way to calculate the sound transmission coefficient of a finite-size thin isotropic panel below half the angular critical frequency, assuming that the panel has a large enough surface density so that the effects of the fluid loading on each side of the panel can be neglected. A rough approximation for the average forced radiation efficiency is given by

$$\langle \sigma \rangle \approx \frac{1}{2} \ln(k_0 \sqrt{S}) \quad (23)$$

Equation (22) is not accurate for small values of $k_0 \sqrt{S}$. In order to include the effect of fluid loading, Wang¹²¹ proposed to compute the diffuse field transmission coefficient, when $k_0 \sqrt{S} > \sqrt{e}$, as

$$\tau_d(\omega) \approx \frac{1}{a^2} \ln \left(\frac{1 + a^2}{1 + \frac{a^2}{k_0 \sqrt{S}}} \right) \quad (24)$$

Because this equation assumes that the panel is limp, it is only valid below about half the critical frequency. At very low values of $k_0 \sqrt{S}$, the azimuthally averaged radiation impedance is

dominated by its imaginary part. According to Thomasson,¹⁹ the azimuthally average normalised radiation impedance for very low values of $k_0\sqrt{S}$ is approximately

$$z_R(\omega, \theta) \approx \frac{(k_0\sqrt{S})^2}{2\pi} + j \frac{k_0\sqrt{S}}{2} \quad (25)$$

Substituting the plate impedance given in equation (20) for the plate impedance given in equation (1), the real part of equation (25) for the real part of the normalised radiation impedance in the numerator of equation (1) and the imaginary part of equation (25) for the normalised radiation impedance in the denominator of equation (1), it is possible to approximate the diffuse sound transmission coefficient as

$$\tau_d(\omega) \approx \frac{4}{\pi} \frac{\rho_0^2 S}{(\rho h + \rho_0\sqrt{S})^2} \quad (26)$$

Wang¹²¹ recommends the use of this equation when $k_0\sqrt{S} < \sqrt{e}$. It should be noted that equation (26) is only accurate for a panel in the frequency range where the panel is behaving like a completely limp panel. The sound transmission coefficient of the panel will be controlled by its stiffness and its boundary conditions below the resonance frequency of its first mode.

When $\omega > \omega_c$, the real part of the azimuthally averaged normalised radiation impedance is much greater than the imaginary part unless the incidence angle θ is close to $\pi/2$ rad, or unless ω_c is a very low frequency. Thus, in the denominator in equation (1), the normalised radiation impedance can be replaced with the real part of the azimuthally averaged radiation impedance, that is, the panel's radiation efficiency. Moreover, replacing the term $1/\cos\theta$ with σ , which is the exact solution for an infinite plate, and is correct for a finite panel except when the incidence angle θ is close to $\pi/2$, equation (1) can be approximated by

$$\tau(\omega, \theta) \approx \frac{\sigma(\theta)^2}{\left[\sigma(\theta) + a\eta r^2 \sin^4\theta\right]^2 + \left[a(1 - r^2 \sin^4\theta)\right]^2} \quad (27)$$

Considering that most of the sound transmission occurs near the peak of sound transmission at the coincidence angle θ_c , an approximation can be made by replacing most of the incidence angles θ , over which the transmission coefficient is integrated to consider a diffuse sound field as given in equation (7), with this angle. Moreover, by introducing the total loss factor: $\eta_{tot} = \eta + 2\eta_{rad}(\theta_c)$, where the radiation loss factor is given by

$$\eta_{rad}(\theta) = \frac{\sigma(\theta)}{2a} \quad (28)$$

it is possible to derive the diffuse field transmission coefficient of a single-leaf panel when the angular frequency is greater than the critical frequency $\omega > \omega_c$

$$\tau_d(\omega) \approx \frac{\sigma(\theta_c)^2}{2a^2 r \eta_{tot}} \left[\arctan \frac{2}{\eta_{tot}} - \arctan \frac{2(1-r)}{\eta_{tot}} \right] \quad (29)$$

This equation was derived by Davy¹²³ as an alternative to Cremer's⁴ formulation

$$\tau_d(\omega) \approx \frac{1}{a^2} \frac{\pi}{2\eta_{tot}} \frac{\sigma(\theta_c)^2}{r} \quad (30)$$

representing the SEA result for resonant transmission. It should be noted that the resonant radiation efficiency is different from the forced radiation efficiency below the critical frequency.

The theoretical results above and below the critical frequency need to be combined. The results derived for above the angular critical frequency can be used on their own, when $\omega > \omega_c$. However, when $\omega < \omega_c$, the result derived for above the angular critical frequency, when the incident angle is $\pi/2$ rad, is added to the result derived for below the critical frequency in order to model the increase in the sound transmission coefficient as the angular critical frequency is approached from below. Brunskog²⁰ has also suggested other ways of combining the two results below the critical frequency. Sewell¹³ suggested multiplying the sound transmission coefficient below the critical frequency by the square of

$$1 - r^2 = 1 - \left(\frac{\omega}{\omega_c} \right)^2 \quad (31)$$

Sharp linearly interpolated the sound reduction index in the decibel – logarithmic frequency domain between half the critical frequency and the critical frequency. According to equation (3), the critical frequency of a panel is inversely proportional to the thickness of the panel. This means that increasing the thickness of the panel to increase its mass does not always increase its sound insulation. Constructing a wall leaf from multiple layers of panels which are only connected to each other by screws, nails or spot gluing at points, which are more than one quarter of the wavelength of freely propagating transverse bending waves in the panels apart, is one way of increasing the mass of a wall leaf without decreasing the critical frequency of the wall leaf. This occurs because the multiple layers can slip relatively to each other when the wall leaf is dynamically bent and hence the bending stiffness does not increase as fast as if the thickness of a single layer wall leaf was increased.

An alternative approach to increase the sound insulation at higher frequencies is to use a double-leaf cavity wall. Below the normal incidence mass-air-mass resonance frequency, a double-leaf cavity wall has the same sound transmission coefficient as a single-leaf wall with the same total mass per unit area as the double-leaf cavity wall. To calculate diffuse field sound transmission coefficient τ_d of double-leaf cavity walls with sound-absorbing material in their wall cavities above the normal incidence mass-air-mass resonance frequency due to transmission via the air in the wall cavity, Sharp¹⁰ multiplied the diffuse field sound transmission coefficients $\tau_{d,1}$ and $\tau_{d,2}$ of the two wall leaves together with an additional factor

$$\tau_d(\omega) = \frac{\tau_{d,1}\tau_{d,2}}{(2H)^2} \quad (32)$$

where

$$H = \begin{cases} k_0 d, & \text{if } k_0 d < 1 \\ 1, & \text{if } k_0 d \geq 1 \end{cases}$$

It should be noted that Sharp obtained the diffuse field sound transmission coefficients of the two wall leaves by calculating the normal incidence sound transmission coefficient of the wall leaf using the mass per unit area of the wall leaf divided by 1.9. In order to include the case when the wall cavity contains little or no sound-absorbing material, Davy replaced H with a sound absorption coefficient α of the wall cavity side surfaces of the two wall leaves. However, he still used the value of H when α was greater than H . For wall cavities containing plenty of sound-absorbing material, Davy¹²³ used a value of α equals 1, while if the wall cavity contained no sound-absorbing material,

he used a value in the range of $0.1 \leq \alpha \leq 0.15$. To also include the size of the wall, Davy followed

Mulholland et al.,⁴⁷ but used the variable limiting angle $\theta_l = \arccos\left(\sqrt{\frac{1}{k_0 \sqrt{S}}}\right)$

$$\tau_d(\omega) = \frac{1 - \cos^2 \theta_l}{(q + p \cos^2 \theta_l)(q + p)} \quad (33)$$

$$q = \frac{1}{2} \left(\frac{a_2}{a_1} + \frac{a_2}{a_2} \right); \quad p = a_1 a_2 \alpha \quad (34)$$

where a_i is computed using equation (18) and α is constrained to be less than or equal to H . equation (33) is only valid below about half the angular critical frequencies of the wall leaves. To extend the range of equation (33) closer to the angular critical frequencies, Davy has added the resonant transmitted sound using a similar technique to that used above for single-leaf walls. Equation (33) overestimates the experimentally measured sound transmission coefficient. To make equation (33) agree better with Sharp's equation (32) which agrees better with experimental results, Davy imposed a maximum value for the limiting angle of 61° for use with double-leaf cavity walls. Above the angular critical frequencies, Davy followed Cremer's approach for single-leaf walls and for double-leaf cavity walls obtained

$$\tau_d(\omega) = \frac{\pi(\eta_1 r_2 + \eta_2 r_1)}{2a_1^2 a_2^2 \eta_1 \eta_2 \alpha^2 \left[(\eta_1 r_2 + \eta_2 r_1)^2 + 4(r_1 - r_2)^2 \right]} \quad (35)$$

where η_i and r_i for i equals 1 or 2 are, respectively, the damping loss factor of the i th wall leaf and the ratio of the angular frequency to the angular critical frequency of the i th wall leaf. Equation (35) assumes that the radiation efficiency above the lower of the two wall leaf critical frequencies is unity and that the product $a_i \eta_i$ is much greater than unity. As in the single-leaf wall case, Davy also obtained formulae for the case when Cremer's extension of the limits of integration over angle of incidence is not made.